Case study 5: Step stool

National 5 · Practical Woodworking · Answers and marking guidance

Award one mark per correct point unless stated. Other valid answers should be credited. Sketches can gain full marks if they show enough detail.

1 (a) Describe how to prepare the square blank before mounting it on the lathe. (2 marks)

- Draw diagonals on each end to find the centres
- Saw along the diagonals on one end for the drive centre / plane the corners off to make it octagonal

1 (b) Describe two safety precautions when using the lathe. (2 marks)

- Wear goggles or a visor
- Tie back hair; no loose clothing or jewellery
- Rotate by hand to check the work clears the tool rest
- Remove the chuck key / tommy bar before switching on

Any 2 points.

2 Name the measuring tool used to check the diameter while turning. (1 mark)

- Outside calipers

3 Explain why beech is suitable for the step. (2 marks)

- Beech is hard and close grained
- So it resists wear and dents from being stood on

4 Name a suitable joint and give one reason for your choice. (2 marks)

- Stopped housing / dowel joint / through housing
- Reason linked to strength, appearance (hidden) or ease of making

5 State one item of PPE that should not be worn when using the lathe, and why. (1 mark)

- Gloves: they can be caught by the rotating work

Total: 10 marks